

Assays for studying *Listeria*-Containing Vacuoles

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Abstract

Listeria monocytogenes is a bacterial pathogen that enters and proliferates in the cytosol of mammalian cells. Following entry, bacteria disrupt the invasion vacuole and reach the cytoplasm, where they replicate and use the actin cytoskeleton to propel themselves from cell to cell [1, 2]. To achieve this cytosolic lifestyle, *Listeria* deploys virulence effectors that hijack diverse cellular processes [3, 4]. Human epithelial cells grown *in vitro* can be used to study the *L. monocytogenes* infectious process. However, rapid multiplication and dissemination of bacteria may induce cell death and detachment, thereby forming lytic plaques. Thus, *in vitro* infections with *L. monocytogenes* have been restricted to short time courses (usually from a few minutes to one day). In order to study *L. monocytogenes* long-term infections, we have set up a protocol, with several modifications to the gentamicin protection assay previously used for short-term infections with this pathogen [5]. In a subset of human cells, such as hepatocytes or trophoblast cells, this protocol enables to observe and study the entrapment of *L. monocytogenes* within vacuoles, termed “*Listeria*-Containing Vacuoles” (LisCVs), after the intercellular dissemination phase of *Listeria*.

Here, this protocol describes the assays used to study LisCVs in a JEG3 trophoblast cell monolayer.

1) Materials

1.1) Long-term infection assays of JEG3 cells with *Listeria monocytogenes*.

- *L. monocytogenes* strain 10403S [6]. Kept in frozen glycerol stock, at -80°C.
- JEG3 trophoblast cells (ATCC HTB-36). Kept in frozen stocks, in liquid nitrogen.
- Bacterial growth medium: Brain Heart Infusion (BHI) medium.
- Cell culture medium: Minimum Essential Medium Eagle (MEM), 10% Fetal Bovine Serum, 1% Non Essential Amino Acids (NEAA), 1mM Sodium Pyruvate (NaP), 2mM L-Glutamine.
- Inoculation and washing cell medium: Minimum Essential Medium Eagle (MEM).
- Bacteria incubator at 37°C.
- Cell incubator with a humidified 10 % CO₂-containing atmosphere at 37 °C.
- 6-well cell culture plates.
- Cell counter.
- Phosphate-buffered saline (PBS).
- Trypsin/EDTA solution.
- Gentamicin.
- 15 mL polystyrene tubes.
- Pipets and pipet tips.

1.2) Quantification of bacterial intracellular loads by Colony Forming Unit (CFU) counts.

- Phosphate-buffered saline (PBS).
- Sterile distilled water kept at 4°C.
- Trypsin/EDTA solution.
- 0.4% Trypan Blue.
- Cell counter.
- BHI agar plates.

- Bacteria incubator at 37°C.

1.3) LisCV staining assays.

- 22 mm square glass coverslips.
- Phosphate-buffered saline (PBS).
- Fixation buffer: 4% Paraformaldehyde in 1X PBS.
- Permeabilization buffer: 0.4% Triton X-100 in 1X PBS.
- Blocking buffer: 2% bovine serum albumin (BSA) in 1X PBS.
- Distilled water.
- Primary antibodies: polyclonal antibodies against *L. monocytogenes*; monoclonal antibodies against LAMP1.
- Secondary antibodies: Alexa Fluor 488-conjugated goat anti-rabbit antibodies; Cy3-conjugated (or Alexa Fluor 555-conjugated) goat anti-mouse antibodies.
- F-actin stain: Alexa Fluor 647-conjugated phalloidin.
- DNA stain: DAPI or Hoechst (1mg/mL stock solution).
- Fine tip forceps.
- Glass slides.
- Mounting medium.
- Inverted motorized fluorescence microscope with 20X (or 40X); 63X (or 100X) objectives.

1.4) Assessment of the integrity of intravacuolar bacteria using the LIVE/DEAD® BacLight(TM) Viability Assay.

- 22 mm square glass coverslips.
- MOPS/MgCl₂ solution (0.1 M 3-(N-morpholino) propanesulfonic acid (MOPS), pH 7.4; 1 mM MgCl₂).
- LIVE/DEAD® BacLight(TM) Bacterial Viability Kit.
- Triton X-100.
- Live/Dead staining and permeabilization solution: mix 1.6 µM SYTO9, 20 µM Propidium Iodide, 0.1% Triton X100 in MOPS/MgCl₂. Briefly mix with a vortex. Keep the solution in the dark.
- Fine tip forceps
- Glass slides
- Clear nail polish.
- Inverted motorized fluorescence microscope with 63X or 100X objective.

2) Safety Warnings

- Containment level: all manipulations with *L. monocytogenes* must be conducted within a laminar flow class II Biological Safety Cabinet (BSC).
- Personal Protective Equipment (PPE): laboratory coat; gloves; safety glasses. Ensure the PPE is maintained in working order and properly used.
- Disinfection: disinfect all work surfaces and materials both prior to and immediately following all work practices and procedures.
- All operations involving Paraformaldehyde must be carried out in a certified chemical fume hood and waste eliminated in a container for hazardous wastes.

3) Experimental procedures

We present the assay in a 6-well plate format, but it can be scaled to 12-well or 24-well plates format.

3.1) Long-term infection assays of JEG3 cells with *Listeria monocytogenes*.

- At least one week before the infection, thaw JEG3 cells by resuspending a frozen cellular aliquot in 10 mL of the cell culture medium in a 15 mL polystyrene tube and centrifuge

during 5 min at $200 \times g$. Discard the supernatant, resuspend the cellular pellet in 10 mL of warm cell culture medium. Add the cells to a 75 cm² flask and transfer it to a cell incubator with a humidified 10 % CO₂-containing atmosphere at 37°C. Subculture cells when they reach 80% confluency.

- At least three days before the infection, and starting from a frozen glycerol stock, streak *L. monocytogenes* strain 10403S on a BHI agar plate and grow bacteria 48h at 37°C (**Note 1**).
- Two (or four) days prior infection, trypsinize a flask of JEG3 cells and seed cells (2mL/well) into 6 well-plates (place coverslips in wells, if you go to step 3.3 or 3.4) in order to reach 90-100% confluency at the onset of infection (**Note 2**). The number of plated wells will vary according to the number of different conditions that will be analyzed (**Note 3**). Prepare a specific plate to count cells at the beginning the infection (3 wells) and at the end of the infection (3 wells). Transfer the plate to the cell incubator, where cells will be allowed to attach and spread for two (or four) days.
- The day before the infection prepare a liquid culture of *L. monocytogenes* by inoculating a bacterial colony from the BHI agar plate to 5 mL of BHI liquid medium in a 15 mL polystyrene tube. Lock the lid and let the culture grow overnight at 37°C in a shaker.
- The day of the infection, wash stationary-phase bacteria in PBS. For this, take 1 mL of the overnight *L. monocytogenes* culture, centrifuge during 2 min at $10,600 \times g$ in a tabletop centrifuge, discard the supernatant, and resuspend the pellet in 1 mL of PBS. Prepare the inoculum by diluting bacteria in the appropriate volume of infection medium (MEM) (**Note 4**). JEG3 cells are infected with 2 mL of inoculum, to achieve a multiplicity of infection (MOI) of approximately 0.1. To check the MOI, make dilutions of the inoculum in PBS, plate them on BHI agar plates and let bacteria grow for 48h at 37°C. Use 3 non-infected wells to count JEG3 cells, after detachment with Trypsin.
- Wash JEG3 cells by aspirating the medium and add 2 mL of MEM to each well.
- Aspirate the medium from JEG3 cells and add 2 mL of the inoculum to each well.
- Centrifuge infected cells 2 min at $300 \times g$ to synchronize the bacterial uptake.
- Transfer the inoculated plate to the cell incubator and allow *L. monocytogenes* to invade cells for 1 h.
- After the cellular infection has been completed for 1 h, gently wash cells with 2 mL per well of pre-warmed MEM. Repeat the washing one more time. Be careful to pipet gently to avoid cell loss (**Note 5**).
- Replace MEM by 2 mL of complete cell culture medium supplemented with 25 µg/mL gentamicin and transfer the 6-well plates back to the cell incubator. Incubate infected cells for 3 days.
- After these 3 days of infection, process cells for CFU counting (see 3.2) or staining assays (see 3.3 and 3.4).

3.2) *Quantification of bacterial intracellular loads by Colony Forming Unit (CFU) counts.*

- After the 3 days of infection, gently aspirate the medium and gently wash the cells with 1mL of pre-warm MEM. Gently remove the medium, and lyse the cells by adding 1mL of cold distilled water per well. Incubate 10 min at 4°C. To completely disrupt the cells, scrape the cells by pressing the pipette tips on the well bottom and rotating while pipetting, and pipette the water 5 times up and down.
- Make a 1:10 dilution by adding 100 µl lysate to 900 µl PBS. Dilute the lysates two more times to obtain 1:100 and 1:1000 dilutions.
- Spread out 50 µl of each lysate dilution on a 10 cm BHI agar plate: 3 dilutions can be plated on one plate divided in three. Let the lysates dry by opening the BHI agar plates under the microbiological hood for a few minutes. Put the lid back on the BHI agar plates, and transfer them to a bacterial incubator. Let bacteria grow for 48h at 37°C.
- Count colonies of each dilution. Use the dilution where colonies are well separated from each other (between 10 and 100 colonies) to calculate the CFU counts.
- To normalize the number of CFU to the number of cell per well, determine the number of viable cells per well by a Trypan Blue assay in the 3 wells that have been kept for this purpose, as follows: remove the medium and detach cells with 250 µl Trypsin/EDTA for 5 min. Resuspend cells with 250 µl of MEM by pipetting up and down (500 µl final). Make sure that suspension is homogeneous. Transfer 20µL aliquot of the cell suspension to a small tube. Add equal 20 µl volume of 0.4%(w/v) Trypan Blue to cell suspension and mix well. This makes a 1:2 dilution (**Note 6**). Use a hemocytometer chamber and count the viable cells (non-blue) and the nonviable cells (blue), according to the instructions provided with the hemocytometer.
- Calculate the ratio of CFU on the number of cell per well.

3.3) *LisCV staining assays.*

- Infect JEG3 cells for 3 days with *L. monocytogenes* in 6-well plates containing coverslips (as in 3.1). Use coverslips that have been previously sterilized 2h at 180°C, in an oven.
- After the 3 days of infection, gently remove the medium and gently wash cells with 1mL PBS.
- Remove PBS with a pipet and fix cells by adding 1mL of the fixation solution (4 % PFA in PBS) to each well. Incubate at room temperature for 30 min under a chemical fume hood.
- Remove the fixative, and gently wash cells with 1mL PBS per well. Renew the washing.
- Permeabilize cells with 1 mL of 0.4% Triton X-100 in 1X PBS, for 4 min.
- Gently wash the coverslips in wells with 2mL PBS, three times.
- Keep coverslips in 1mL blocking solution (PBS with 2% BSA) per well.

- Prepare the primary antibodies solution by diluting rabbit polyclonal *L. monocytogenes* antibodies and mouse LAMP1 monoclonal antibodies in PBS with 2% BSA. Maintain the antibody solution on ice.
- Prepare an incubation chamber, with a wet Whatman paper and a parafilm on it (firmly stick to the Whatman and without any bubble). Mark the parafilm with a number to correspond to a coverslip (**Note 7**).
- Dispatch drops of 40 µl of the antibody solution on the parafilm, for each coverslip. Use a fine tip forceps to pick up the coverslip and return the coverlip on the primary antibody (i.e. the cell face on the antibody solution), and incubate for 60 min.
- Prepare the secondary antibody solution by diluting the secondary AlexaFluor 488-conjugated anti-rabbit antibody (1:400), and Cy3-conjugated (or Alexa Fluor 555-conjugated) anti-mouse antibody (1:400), DAPI or Hoechst (1:1000) and Alexa Fluor 647-conjugated phalloidin (1:400) in PBS supplemented with 2% BSA (**Note 8**).
- Gently detach each coverslip by injecting PBS under it. Wash the coverslip in three successive bathes of PBS.
- Place the coverslip (cell side down) on a drop of 40 µl of the secondary antibody solution dispatched on a parafilm in the incubation chamber. Incubate for 30 min at room temperature in the dark to protect the fluorescent probes.
- Gently detach each coverslip by injecting PBS under it. Wash the coverslip in two successive bathes of PBS and one bath of distilled water.
- Remove excess water by touching the edge of the coverslip to a paper towel and mount the coverslip with 10 µL mounting medium onto a glass slide. Avoid forming bubbles.
- Let the coverslip dry overnight at room temperature, in the dark to protect the fluorescent probes.
- Acquire images in the 350 nm (DAPI/Hoechst, to visualize nuclei and bacterial DNA), 488 nm (*L. monocytogenes*), 546 nm (LAMP1), and 647 nm (F-actin) channels on an inverted motorized fluorescence microscope, equipped with a digital camera. Use a 40X (or 20X) objective to observe bacterial spreading in the cell monolayer, and 100X (or 63X) objectives to observe Listeria-Containing Vacuoles (LisCVs). LisCVs appear as perinuclear LAMP1-positive compartments containing F-actin-negative *L. monocytogenes*.

3.4) Assessment of the integrity of intravacuolar bacteria using the LIVE/DEAD® BacLight(TM) Viability Assay

We adapted the Live/Dead BacLight Bacterial Viability assay from references [7] and [8], to assess whether bacteria in LisCVs have intact or damaged membranes. This assay is based on the discriminative labeling of two fluorescent nucleic acid-stains: the green membrane-permeant dye SYTO9 and the red membrane-impermeant dye propidium iodide (PI), which only crosses damaged cell membranes. Both dyes penetrate dead cells, but SYTO9 fluorescence is reduced in presence of PI. As such, intact bacteria appear green, while those with damaged membranes appear red.

To allow the dyes to reach vacuolar *Listeria*, this staining protocol is performed in the presence of 0.1% Triton X100, which permeabilizes host cell membranes, including vacuoles containing bacteria [7]. The efficiency of permeabilization is assessed by the bright staining of host cell nuclei with PI, as the dye crosses the double-membrane of the nuclear envelope.

- Infect JEG3 cells for 3 days with *L. monocytogenes* in 6-well plates containing coverslips (as in 3.1). Then process each coverslip separately.
- Gently wash cells twice with 1 mL of MOPS/MgCl₂ solution (0.1 M 3-(N-morpholino) propanesulfonic acid (MOPS), pH 7.4; 1 mM MgCl₂).
- Add 1 mL of the Live/Dead staining and permeabilizing solution (1.6 µM SYTO9, 20 µM Propidium Iodide, 0.1% Triton X100 in MOPS/MgCl₂) to the well.
- Incubate 10 min with centrifugation at 300g to prevent cell detachment and 5 min at room temperature in the dark.
- Gently wash cells twice with 1 mL in the MOPS/MgCl₂ solution (use a pipet to aspirate the buffer).
- Mount the coverslip onto glass slides and seal with clear nail polish. Incubate 2 min at 37°C.
- Immediately examined the mounted coverslip under the microscope (63X or 100X objective) for a maximum of 10 minutes. Bacteria appear as tiny green or red filaments (as the dyes stain the bacterial nucleoid) (**Note 10**).

4) Notes

- **Note 1.** The plate can be stored at 4°C for 15 days.
- **Note 2.** Dilute cells to achieve between 1×10^5 and 2×10^5 cells/well when cells are plated two days prior infection, and between 1×10^4 and 3×10^4 cells/well when cells are plated four days prior infection. Cell quantification may vary from one laboratory to another because of the use of different cell counter systems, thus the experiment has to be adapted in order to achieve 90-100% confluency at the onset of infection.
- **Note 3.** We typically seed cells in triplicate (three independent wells) for each condition.
- **Note 4.** The specific MOI of the inoculum needs to be adjusted according to the *L. monocytogenes* strain that is examined in the assay: when using strain 10403S, we prepare an inoculum containing between 4×10^4 and 8×10^4 bacteria in 2 mL (the final MOI is 0.1, estimated for 4×10^5 and 8×10^5 cells per well).
- **Note 5.** When using MOI above 0.1 (e.g. 1-10), expose cells 10-min with 2mL of MEM containing gentamicin 100 µg/mL to rapidly kill extracellular bacteria. This treatment prevents the lytic action of the LLO toxin, which is secreted by *L. monocytogenes*. After this treatment, wash cells with 2 mL MEM and add complete cell medium with gentamicin 25 µg/mL.
- **Note 6.** Do not keep cells in Trypan Blue solution for more than 10 minutes before cell counting (after 10-15min in Trypan Blue, cells start dying and the count will not be accurate).
- **Note 7.** Never let the coverslips go dry. Check that the Whatman paper stays wet, with distilled water.

- **Note 8.** Check the excitation and emission filters of the microscope to avoid overlaps of fluorescence signals. Adjust the secondary antibodies accordingly.
- **Note 9.** To detect bacteria with SYTO9 and PI, one has to use a high intensity of light, which can become cytotoxic for bacteria. The examination of Live/Dead intracellular bacteria has therefore to be performed within 5 to 10 minutes. A field of infected cells needs to be found rapidly. The success of an experiment is thus function of the efficiency of infection.
- **Note 10.** It is difficult, and quite subjective, to count in vacuoles the number of bacterial filaments stained by the BacLight probes. For this reason, this is a qualitative assay.

5) References

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